INCH-POUND
MIL-T-49447(CR)
2 November 1988

MILITARY SPECIFICATION

TEMPERATURE SENSOR, RESISTANCE

This specification is approved for use within the Communications-Electronics Command, Fort Monmouth, New Jersey, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of temperature sensitive electrical resistance sensor which is used for measuring atmospheric temperature.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder: Lead Alloy, Tin-Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire Solid Form

MILITARY

MIL-P-116 - Preservation Methods of.
MIL-M-13231 - Marking of Electronic Items.
MIL-F-14256 - Flux, Soldering, Liquid (Rosin Base).
MIL-E-55585 -Electronics Equipment and Parts, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC - N/A FSC 6660
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STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-726 - Packaging Requirements Code.

(Copies of specifications, standards, drawings and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supercede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish first article units for first article inspection and approval (see 4.3 and 6.2).

3.2 Construction. The Temperature Sensor, Resistance shall be constructed in accordance with the requirements of this specification. The Temperature Sensor shall consist of a thin rod of semiconductive ceramic material, coated with a material which is highly reflective to solar radiation. The resistance of this sensor shall be a function of the temperature of the sensor.

3.2.1 Dimensions. The temperature sensor shall have a maximum coated diameter of 0.042 inches (1.07 mm) and a length of 1.5 ± 0.3 inches (38.1 ± 7.6 mm).

3.2.2 Terminals. The temperature sensor shall have two straight conducting leads of hard-drawn, tinned, copper wire. These leads shall be 1.5 ± 0.2 inches (38.1 ± 5.1 mm) in length, and 0.01 ± 0.002 inch (0.254 ± 0.051 mm) in diameter. The maximum diameter of solder connection connecting the leads to the ends of the ceramic rod shall be 0.08 inch (2.032 mm), and the maximum length of the connection shall be 0.25 inch (6.35 mm), excluding feathering out of the solder.

3.2.3 Reflective coating. The reflective coating shall conform to the requirements of 3.7.1.

3.2.4 Container. The sensor shall be individually sealed in glass or plastic containers. The maximum width of inner diameter of these containers

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shall be 0.4 inch (10.2 mm), and the maximum inner length shall be 3 inches (76.2 mm). The sensor shall be protected from shock and vibration within this container. The containers shall be capable of being opened readily without the use of special tools. The Temperature Sensor within the sealed container shall be capable of withstanding the test of 4.9.1.

Alternatively, the sensors may be supplied in individually compartmentalized boxes. Each sensor shall be wrapped in a protective label wrapper and inserted into its compartment. Each box shall be labeled with the manufacturer's name, the lot number, quantity of sensors enclosed, and the lock-in resistance value of the elements.

3.3 Parts, materials, and processes.

3.3.1 Soldering.

3.3.1.1 Solder. Solder shall conform to specification QQ-S-571. For soldering electrical connections, composition SN60, type RA or S, shall be used. When type S solder is used for soldering electrical connections, the flux shall conform to specification MIL-F-14256 activated.

3.3.1.2 Acid or acid salts. No acid or acid salts shall be used in preparation for, or during soldering; however, exception is permitted for preliminary tinning of electrical connections. Where acid or acid salts are used, as permitted above, all residue shall be completely neutralized and removed immediately after use.

3.3.1.3 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections.

3.3.2 Cleaning. After the leads have been attached to the ceramic rod, and before the dipping process is commenced, the sensor shall be cleaned in accordance with the best commercial practice. The cleaning process shall have no deleterious effect on the element or on the bonded connections. All traces of corrosive agents shall be removed.

3.3.3 Waterproofing treatment. Upon completion of the reflective coating process, the elements shall be dipped in a waterproofing solution and allowed to dry. The waterproofing solution shall consist of one part of Number 200 Fluid, as manufactured by the Dow Corning Corp., viscosity 200 centistrokes at 25°C, or equal, and two parts of carbon tetrachloride by volume.

3.3.4 Recycled, virgin, and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

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3.4 Marking. The sensor nomenclature, order number and name of manufacturer shall be printed on an appropriately sized sheet of white bond paper and inserted in the vial of the container in such a manner that this information is visible from outside of the container, or the required marking shall be stamped on the outside surface of the container, either marking shall conform to MIL-M-13231.

3.5 Safety engineering.

3.5.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.8).

3.5.2 Corner and edge rounding. All exposed corners and edges shall be rounded where appropriate to minimize personnel hazards.

3.5.3 Radioactive materials. Radioactive materials shall not be used (e.g.luminous dials/marking, electron tubes, surge arrestors and lenses).

3.6 Weight. The weight of Temperature Sensor, excluding container, shall not exceed 9 grams (see 4.6).

3.7 Performance requirements. The Temperature Sensor shall meet the following performance requirements:

3.7.1 Reflectivity. The weighted mean reflectivity shall be not less than 92 percent when tested in accordance with 4.7.2.

3.7.2 Humidity resistance. The change of resistance of the sensor shall be less than 70 ohms when measured at 20 percent and 100 percent relative humidity at a temperature of 30°C ±3°C. (see 4.7.3).

3.7.3 Polarization. When the sensor is subjected to the polarizing current and conditions specified in 4.7.4, the resistance of the sensor shall not have changed by more than 70 ohms (0.5 percent). (see 4.7.4)

3.7.4 Thermal shock. The sensor shall be capable of sustaining abrupt changes in temperature of over 100°C without physical or electrical damage. (see 4.7.5).

3.7.5 Mechanical strength. When tested in accordance with 4.7.6, the elements shall support a load of at least 100 grams without physical or electrical damage.

3.7.6 Resistance at 30°C. The resistance of the temperature sensor at 30°C shall be 14,000 ohms ±6.0 percent. The data shall be provided with each sensor with an accuracy of 0.2°C. (see 4.7.7).

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3.7.7 Resistance-temperature characteristic. The resistance-temperature ratio shall be defined as the ratio of the resistance of the sensor at any given temperature to the resistance of the sensor at 30°C. The temperature in degrees Celcius as a function of this ratio is described by Equation 1. The resistance temperature ratio shall follow this Equation 1 within ±0.5°C (table V is a listing of these ratios).

Equation 1

$$t = \left[\frac{1}{\sum_{k=0}^{3} A_k (\ln(R_t/R_{30}))^k} \right] - 273.15$$

t = The measured temperature (degree Celsius).
R_t = The temperature sensor resistance (at any t°C).
R_{30} = The temperature sensor lock-in resistance at +30°C.

A_0 = 3.2987 E-03
A_1 = 4.7764 E-04
A_2 = 3.0029 E-06
A_3 = 1.5108 E-06

3.8 Workmanship. The equipment shall be manufactured and assembled in accordance with requirements 5, 7 and 9 of MIL-STD-454 and the applicable portions of the following paragraphs herein.

3.2.1	Dimensions	3.3.1	Soldering
3.2.2	Terminals	3.3.2	Cleaning
3.2.3	Reflective coating	3.3.3	Waterproofing treatment
3.2.4	Container	3.4	Marking

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). (Does not include packaging).
b. Inspections covered by subsidiary documents (see 4.4).
c.
1. Quality conformance inspection of equipment before packaging (see 4.5).
2. Packaging inspection (see 4.9).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish thirty first article units of the Temperature Sensor for group A, B, and C testing.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in table I, and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test reports shall be as required in the contract.

TABLE I. First article inspection.

Inspections	Reqt para	Test para	Order of tests (30 units)
1. Inspections covered by subsidiary documents 1/	3.3 3.4	4.4	Inspection to be performed on all units
2. Group A inspection 1/	See table II		Tests to be performed on all units
3. Group B inspection 1/	See table III		Tests to be performed on all units
4. Group C inspection Thermal shock Humidity resistance Polarization	 3.7.4 3.7.2 3.7.3	 4.7.5 4.7.3 4.7.4	 1 2 3

1/ The inspection 1 to 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable documents as part of the inspection required by this specification, and the inspection requirement specified in the contract.

Item	Where required
Parts, materials and processes	3.3
Marking	3.4

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, and 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspection in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection level II and acceptable quality levels indicated in table II.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the government.

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TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	AQL major 2/	AQL minor 2/
Visual and mechanical	3.8	4.8	1%	4%
Performance	3.7	4.7	1%	4%
Resistance at 30%	3.7.6	4.7.7	1%	1/
Resistance-temperature characteristic	3.7.7	4.7.8	1%	1/

1/ All electrical defects are major.
2/ Defects per hundred units (dphu).

4.5.1.2 Classification of defects. The defects of group A inspection shall be classified as follows:

a. Visual and mechanical defects see 4.8.
b. Resistance at 30°C:
Major - Resistance at +30°C exceeds 14,000 ohms ±9 percent and lock-in resistance value exceeds a temperature equivalent of 0.3°C.
Minor - Resistance at +30°C exceeds 14,000 ohms ±6 percent, but not more than 14,000 ohms ±9 percent and lock-in resistance value does not exceed a temperature equivalent of 0.3°C.

c. Baseline resistance:
Major - Lock-in resistance error exceeds a temperature equivalent of 0.3°C.
Minor - Lock-in resistance error exceeds a temperature equivalent of 0.2°C but but not more than 0.3°C.

d. Resistance - temperature characteristics:
Major - Calibration errors exceed limits by more than ±0.2°C.
Minor - Calibration errors exceed limits, but not by more than ±0.2°C.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection, and on samples selected from units that have been subjected to, and met the group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The group B sample plans, for the AQL's listed in table III, shall be as follows:

Inspection level AQL

S-3 1.5 %
S-4 6.5 %

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4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in an order satisfactory to the government.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test paragraph	AQL
Weight	3.6	4.6	1.5%
Reflectivity	3.7.1	4.7.2	1.5%
Mechanical strength	3.7.5	4.7.6	6.5%

4.5.2.3 Classification of defects. The defects of group B inspection shall be classified as follows:

Mechanical strength:
Major - Temperature sensor breaks.
Minor - N/A.

Reflectivity:
Major - Reflectivity is less than 90 percent.
Minor - Reflectivity is less than 92 percent but not less than 90 percent.

4.5.3 Group C inspection. Group C inspection shall be performed on inspection lots that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1 .

4.5.3.1 Sampling for inspection. Thirty sensors shall be selected at random from the first 500 produced and subjected to the tests of table IV. Thirty sensor samples from each 10,000 units produced thereafter shall be subjected to the tests of table IV.

4.5.3.2 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to, and pass group A inspection only.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Thermal shock	3.7.4	4.7.5
Humidity resistance	3.7.2	4.7.3
Polarization	3.7.3	4.7.4

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4.6 Weight. The Temperature Sensor shall be weighed on a calibrated scale to determine compliance with the weight requirement of 3.6.

4.7 Performance testing. The Temperature Sensor shall be tested for compliance with specified performance requirements of 3.7. The measurements shall be performed in accordance with established inspection standard specified in 4.7.1.

4.7.1 Inspection standards.

4.7.1.1 Temperature standard. The contractor shall provide a standard of temperature measurement in the form of a precision resistance thermometer or equal, for standarizing the calibration and inspection equipment used in the manufacture of the temperature sensor. The standard shall cover the full operating range of the sensor, and shall provide an accuracy in temperature to within ±0.05°C. Copies of certificates of accuracy for the standard used, issued by a competent authority within one year prior to each time it is used on the contract shall be furnished to the contracting officer.

4.7.1.2 Temperature measurements. The accuracy of all temperature measuring equipment used for the calibration and inspection of the temperature sensor shall be maintained at all times to within ±0.05°C of the standard. This measuring equipment shall be calibrated against the standard specified in 4.7.1.1 above, before the beginning of the inspection, and at regular monthly intervals thereafter.

4.7.1.3 Temperature baths. Temperature baths shall be of a neutral fluid that will not affect the calibration of the temperature sensor tested. The baths shall be thoroughly stirred so that any temperature gradients between the temperature measuring equipment and the elements under test do not exceed 0.05°C. The drift of the temperature of the baths shall not exceed 0.1°C per minute. The bath shall be provided for temperatures covering the range from +40°C to -60°C. The resistance of the temperature bath, measured between two electrodes spaced a distance equal to the length of the body of the temperature sensor, shall be maintained at a value of at least 500 times of the resistance of the sensor at the testing temperature, under all testing conditions.

4.7.1.4 Temperature chambers. If an air testing chamber is to be used, the air in the chamber shall be thoroughly mixed and circulated over the sensor being tested. Temperature gradients within the measuring space in the chamber shall not exceed 0.5°C per minute. The chamber shall be suitable for operating at any point in the temperature range from +65°C to -70°C.

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4.7.1.5 Resistance measurements. Resistance measuring equipment shall be available in the form of a precision Wheatstone Bridge, electronic bridge, precision potentiometer, or by equivalent means of measuring the resistance of the temperature elements at any temperature in the range of the element to any accuracy with ±0.1 percent of the resistance measured.

4.7.2 Reflectivity test. The weighted mean reflectivity shall be computed using the formula:

r = 100(1-0.713 Rw/Rb).
r = the weighted means reflectivity.
Rw = resistivity of the white element (temperature sensor).
Rb = resistivity of the black element.

The procedure to determine the resistivity of the white or black element is described below:

With the white test element in the test position, and the light bulb extinguished, the potentiometer in the bridge shall be adjusted so that both the galvanometer and the decade box read zero. The light bulb shall then be turned on, and the decade box shall be adjusted to retain the zero galvanometer reading. When the galvanometer ceases to show additional deflection, due to radiation heating of the test element, the value of resistance on the decade box shall be recorded. This shall be R_W.

Repeat the same procedure for black element to determine Rb.

Weighted mean reflectivity values of 92 percent or greater shall meet the requirements of 3.7.1.

4.7.3 Humidity resistance. The temperature sensor shall be placed in a temperature controlled chamber, at a temperature of 30°C ±10°C and the resistance of the element measured at 20 ± 10 percent and 90 +5, -0 percent relative humidity to determine the requirement of 3.7.2.

4.7.4 Polarization. During this entire test the temperature of the chamber shall be maintained at 25°C ±5°C. The sensor shall remain in the chamber until its resistance has attained equilibrium. The resistance shall then be measured. A potential of 9 volts DC shall then be applied, through a series 50,000 ohms resistor, to the element for one half hour. Then the resistance of the sensor shall again be measured. The resistance shall not have changed more than 70 ohms from the previously measured resistance, after correction for any temperature drift in the chamber. (see 3.7.3).

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4.7.5 Thermal shock. The temperature sensor shall be placed in a temperature controlled chamber or bath at a temperature not warmer than -70°C, for a period of two minutes, following which the sensor shall be removed and immediately placed in a temperature chamber or bath at a temperature not colder than +60°C for two minutes. The element shall then be examined for physical and electrical damage, such as resistance ratio does not conform to table V (see 4.7.8), loose terminal leads, cracked coating or ceramic rod, or peeled coating.

4.7.6 Mechanical strength. The body of the temperature sensor shall be supported on two knife edges spaced one inch apart. A knife edge load of 100 grams shall be applied to the body of the element at a point approximately equidistant from each supporting knife edge. There shall be no breakage or cracking of the element. (see 3.7.5).

4.7.7 Resistance at 30°C. The resistance of the sensor at the 30°C (+5°C, -2°C) shall be given by Equation 2, valid between 0°C and 50°C:

Equation 2

$$R_{30} = R_t \exp \left[\sum_{k=0}^{3} B_k t^k \right]$$

B_0 = -7.535 E-01
B_1 = 2.7637 E-02
B_2 = -8.9869 E-05
B_3 = 1.978 E-07

R_t = the temperature sensor resistance (at any t°C).

t = Ambient temperature.

Eg.	t°C	R_t (ohms)	R_{30} (ohms)
	28.00	14742	14082
	30.00	13920	13921
	35.00	12531	14019

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4.7.8 Resistance-temperature ratio characteristic. The resistance of the temperature sensor shall be measured in a temperature bath or temperature chamber. Resistance measurements shall be made at each of the following temperature points:

a. +30°C, (+5°C, -1°C)
b. 0°C, (+2°C, -5°C)
c. -20°C, (+2°C, -5°C)
d. -40°C, (+2°C, -5°C)
e. -60°C, (+2°C, -5°C)

The indicated sensor temperature shall be determined by using Equation 1 (3.7.7). The difference between this indicated temperature and the bath temperature is the transfer characteristic accuracy. The accuracy shall conform to the requirements of 3.7.7.

TABLE V. Resistance-temperature ratio. 2/

Temperature °C	Nominal Ratio
+50	0.652
+40	0.802
+30	1.000
+20	1.265
+10	1.626
0	2.124
-10	2.827
-20	3.838
-30	5.322
-40	7.551
-50	10.980
-60	16.400
-70	25.2 1/
-80	40.0 1/
-90	66.0 1/

1/ These values are nominal values, furnished for information only.
2/ The resistance-temperature ratio is defined by the following relationship:

$$\text{Resistance - Temperature} = \frac{\text{Resistance of element at given temperature.}}{\text{Resistance of element at } 30^{\circ}\text{C.}}$$

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4.8 Visual and mechanical inspection. The Temperature Sensor shall be examined for the following defects:

Major defects:
- Container dimensions not within specification limits.
- Sensor dimensions outside of specified tolerances.
- Terminal dimensions outside of specification limits.
- Conducting leads not tinned.
- Improper wrap, less than 1/2 turn.
- Unsoldered joint.
- Insufficient solder.
- Minimum bridge of solder; less than twice the diameter of the wire or less than 3/32 inch, whichever is greater.
- Cold solder joint, chalky appearance, lacks metallic luster, presents rough "piled up" appearance, movement of wire or solder upon pick application.
- Rosin joint, presence of excessive rosin.

Minor defects:
- Improper wrap, 1/2 turn or more but less than one turn.

In addition, Temperature Sensor shall be visually inspected to determine conformance to the requirements of 3.4 and 3.5.

4.9 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.9.1 First article inspection of packaging. Unless otherwise specified in the contract, First Article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.9.2 Quality conformance inspection of packaging.

4.9.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.9.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.9.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table VI, Packing inspection provisions. Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

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TABLE VI. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 10-1-1-00-00-EA-X-B7-Q-ED-A.

5.1.2 Level B. 10-1-1-00-00-EA-X-B7-Q-ED-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The Temperature Sensor, Resistance, is used as a temperature sensor with Radiosonde Set ML-659(V).

6.2 Order data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendments thereto.
b. Level A or level B preservation and packing (see section 5).
c. When first article inspection rough handling test(s) are not required.
d. The contractor should apply for nomenclature in accordance with the applicable clause in contract.
e. Marking and shipping of samples.
f. Place of final inspection.
g. Number of first articles to be submitted for approval (see 3.1).
h. Number of first article packaging inspection test reports require Acquisition Activity approval prior to production unit packing.

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6.3 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.5 Subject Terms Key Word) List

Temperature
Sensor
Temperature Sensor Resistance Sensor
Resistance Material-semiconductor ceramic
Radiosmode Set ML-659(V)

Custodian:
Army - CR

Preparing activity:
Army - CR
Project 6660-A079

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-T-49447(CR)	Temperature Sensor, Resistance

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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